The Krebs Cycle

also called the citric acid cycle or the tricarboxylic acid (TCA) cycle — the hub where all fuels converge

DEFINITION

A closed loop of **eight enzyme-catalysed reactions** that completely oxidises the acetyl group of acetyl-CoA to **two molecules of CO_2**, capturing the released electrons as **NADH** and **$FADH_2$** and one high-energy phosphate as **GTP**.

The cycle rebuilds its own starting material, oxaloacetate — so it acts catalytically and is never used up. One oxaloacetate can shuttle an unlimited number of acetyl groups to CO_2.

KEY FACTS

- **Location** mitochondrial matrix — except succinate dehydrogenase, embedded in the inner membrane (cytosol in prokaryotes)
- **Turns per glucose** 2
- **8 steps** · 4 oxidations · 2 decarboxylations · 1 substrate-level phosphorylation
- **Described by** Hans Krebs, 1937 — Nobel Prize 1953

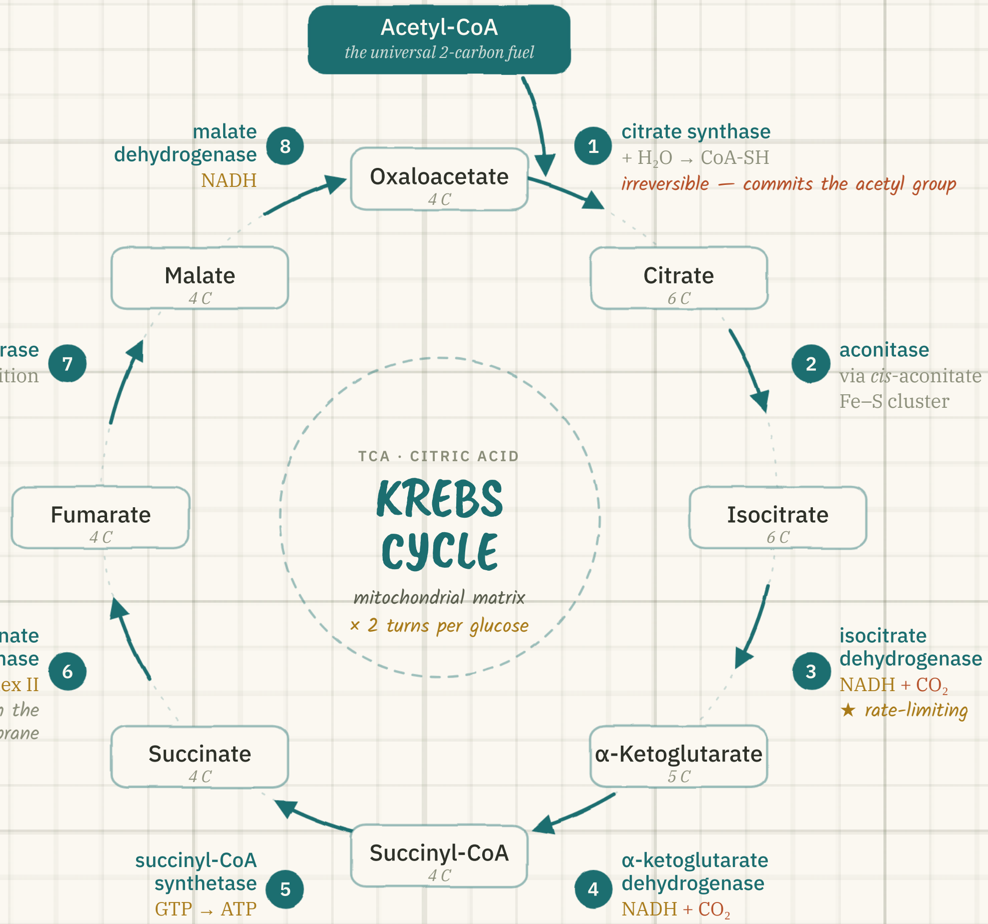
NET EQUATION — ONE TURN

$$\text{Acetyl-CoA} + 3\,NAD^+ + FAD + GDP + P_i + 2\,H_2O$$
$$\rightarrow 2\,CO_2 + 3\,NADH + 3\,H^+ + FADH_2 + GTP + \text{CoA-SH}$$

COMMON MISREADING

*The cycle **never touches O_2**. Oxygen is consumed downstream at Complex IV. The cycle is aerobic by dependence, not by chemistry — without the electron transport chain to regenerate NAD^+ and FAD, it stalls within seconds.*

The wheel



Eight steps, four oxidations, two carbons in and two carbons out — the wheel turns twice for every glucose.

Ledger for one turn

3	1	1	2	8	10
NADH	$FADH_2$	GTP	CO_2	E⁻ HARVESTED	ATP EQUIV.

↳ 3 NADH × 2.5 + 1 $FADH_2$ × 1.5 + 1 GTP = **10 ATP per turn**, so **20 ATP per glucose** come from the cycle alone.

Step zero — the link reaction

NOT PART OF THE CYCLE — BUT THE GATE INTO IT

Pyruvate (3 C · from glycolysis) → **Pyruvate dehydrogenase complex** (E1 · E2 · E3) → **Acetyl-CoA** (2 C · enters the wheel)

releases $-\ CO_2$ $+\ NADH + H^+$

Irreversible — this is the point of no return. Once pyruvate is decarboxylated, its carbons can never go back to glucose, which is why fatty acids cannot be converted to glucose in humans.

Five cofactors, all needed:

Thiamine pyrophosphate · Lipoamide · CoA-SH · FAD · NAD^+

"Tender Loving Care For Nancy"

The eight steps

#	REACTION	ENZYME	CHEMISTRY	YIELD	ΔG°′
1	Acetyl-CoA + oxaloacetate + H_2O → **citrate** + CoA-SH	**citrate synthase**	Claisen condensation; hydrolysis of the thioester bond supplies the driving force	—	**−32.2**
2	Citrate ⇌ *cis*-aconitate ⇌ **isocitrate**	**aconitase**	Dehydration then rehydration on the opposite face; Fe–S cluster. Moves the –OH onto a carbon that *can* be oxidised	—	+13.3
3	Isocitrate + NAD^+ → **α-ketoglutarate** + CO_2	**isocitrate dehydrogenase**	Oxidative decarboxylation via an oxalosuccinate intermediate — **rate-limiting step**	**NADH · CO_2**	**−20.9**
4	α-Ketoglutarate + NAD^+ + CoA-SH → **succinyl-CoA** + CO_2	**α-ketoglutarate dehydrogenase complex**	Oxidative decarboxylation; structurally and mechanistically a twin of the PDH complex — same five cofactors	**NADH · CO_2**	**−33.5**
5	Succinyl-CoA + GDP + P_i → **succinate** + GTP + CoA-SH	**succinyl-CoA synthetase** (succinate thiokinase)	The cycle's only **substrate-level phosphorylation** — thioester energy is captured directly as a nucleotide triphosphate	**GTP**	−2.9
6	Succinate + FAD → **fumarate** + $FADH_2$	**succinate dehydrogenase** (= Complex II)	FAD is used because the C–C dehydrogenation is not energetic enough to reduce NAD^+; electrons pass straight to ubiquinone	**$FADH_2$**	≈ 0
7	Fumarate + H_2O → **L-malate**	**fumarase** (fumarate hydratase)	Stereospecific *trans* hydration — produces only the L isomer, never D	—	−3.8
8	L-Malate + NAD^+ → **oxaloacetate** + NADH + H^+	**malate dehydrogenase**	Alcohol oxidised to a ketone, regenerating the acceptor and closing the loop	**NADH**	+29.7

*$\Delta G°'$ in kJ·mol⁻¹ under standard conditions (Lehninger values; textbooks vary by a few kJ). Steps **1, 3 and 4** are strongly exergonic and effectively irreversible — they are the cycle's control points.*

WHY STEP 8 RUNS UPHILL

Malate dehydrogenase has $\Delta G°' = $ **+29.7 kJ·mol^{-1}** — thermodynamically it should run backwards. It doesn't, because citrate synthase consumes oxaloacetate the instant it appears, holding matrix [OAA] below ~1 μM.

The cycle is pulled forward by product removal, not pushed by each step. Coupling, not brute force.

ORDER MNEMONIC

***C**itrate **I**s **K**rebs' **S**tarting **S**ubstrate **F**or **M**aking **O**xaloacetate*

Citrate → **I**socitrate → α-**K**etoglutarate → **S**uccinyl-CoA → **S**uccinate → **F**umarate → **M**alate → **O**xaloacetate

Following the carbons

4 C + 2 C → 6 C → 6 C → − CO_2 5 C → − CO_2 4 C

↳ *oxaloacetate + acetyl-CoA → citrate → isocitrate → α-ketoglutarate → succinyl-CoA, then **four more four-carbon steps** rebuild oxaloacetate without further carbon loss.*

THE SUBTLETY EXAMINERS LOVE

The two CO_2 released in a given turn are **not** the two carbons that just arrived as acetyl-CoA. Isotope labelling shows both released carbons originate from the **oxaloacetate** already in the cycle: the C4 of oxaloacetate becomes C1 of α-ketoglutarate and leaves at step 4. The acetyl carbons are retained in the four-carbon skeleton and leave on *later* turns.

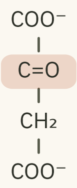
↳ *Carbon bookkeeping balances every turn (2 in, 2 out) — but the individual atoms do not.*

Nine molecules

Every intermediate is a short carbon chain capped with carboxylates. Only one group changes at each step — follow the highlight and the whole cycle becomes a single running edit to one molecule.

hydroxyl —OH · carbonyl C=O · thioester —S-CoA · double bond C=C

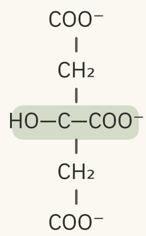
1 Oxaloacetate *4 C*



KETONE

the acceptor — regenerated every turn

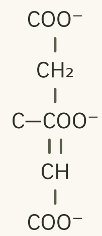
2 Citrate *6 C*



3° ALCOHOL

tertiary alcohol — cannot be oxidised

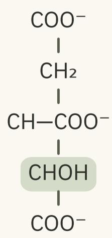
3 cis-Aconitate *6 C*



ENZYME-BOUND

the bound intermediate of aconitase

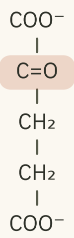
4 Isocitrate *6 C*



2° ALCOHOL

secondary alcohol — now oxidisable

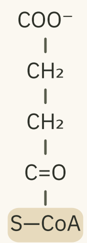
5 α-Ketoglutarate *5 C*



A-KETO ACID

first CO_2 has left

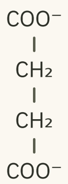
6 Succinyl-CoA *4 C*



THIOESTER

thioester — the energy store

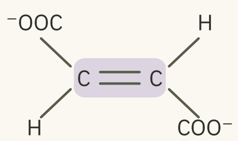
7 Succinate *4 C*



SYMMETRIC

symmetric — no chiral centre left

8 Fumarate *4 C*

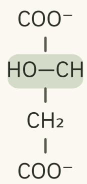


trans (E)

ALKENE

trans only — maleate is not a substrate

9 L-Malate *4 C*

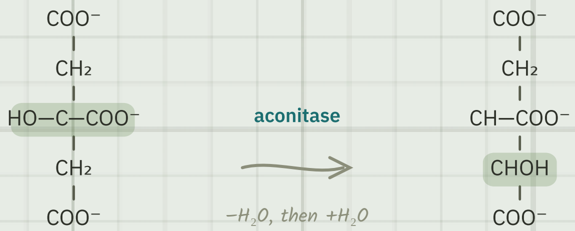


L ONLY

water added across the double bond

Why step 2 exists

AN ENTIRE STEP TO MOVE ONE HYDROXYL



3° ALCOHOL
citrate

2° ALCOHOL
isocitrate

Citrate is a dead end. Its hydroxyl sits on a carbon carrying **no hydrogen** — a tertiary alcohol — and you cannot oxidise an alcohol to a carbonyl without a hydrogen to remove.

Aconitase dehydrates citrate to *cis*-aconitate and rehydrates it the other way round, relocating the –OH to the neighbouring carbon. Isocitrate is a **secondary** alcohol with a hydrogen to give up, and step 3 can proceed.

↳ The cycle spends a whole enzyme on a rearrangement that changes no bonds' count — only their position.

OGSTON'S PUZZLE · 1948

A symmetric molecule that behaves asymmetrically

Citrate is achiral and looks perfectly symmetric — two identical $-CH_2COO^-$ arms. Yet labelling shows the cycle always attacks *the same arm*, the one contributed by oxaloacetate.

Alexander Ogston resolved it: an enzyme binding a **prochiral** substrate at **three points** can tell two apparently identical groups apart, because only one orientation fits the site. Citrate stops being symmetric the moment it is held.

↳ This is why the carbons leaving as CO_2 are predictable, not random — see sheet 2.

WHY THIOESTERS

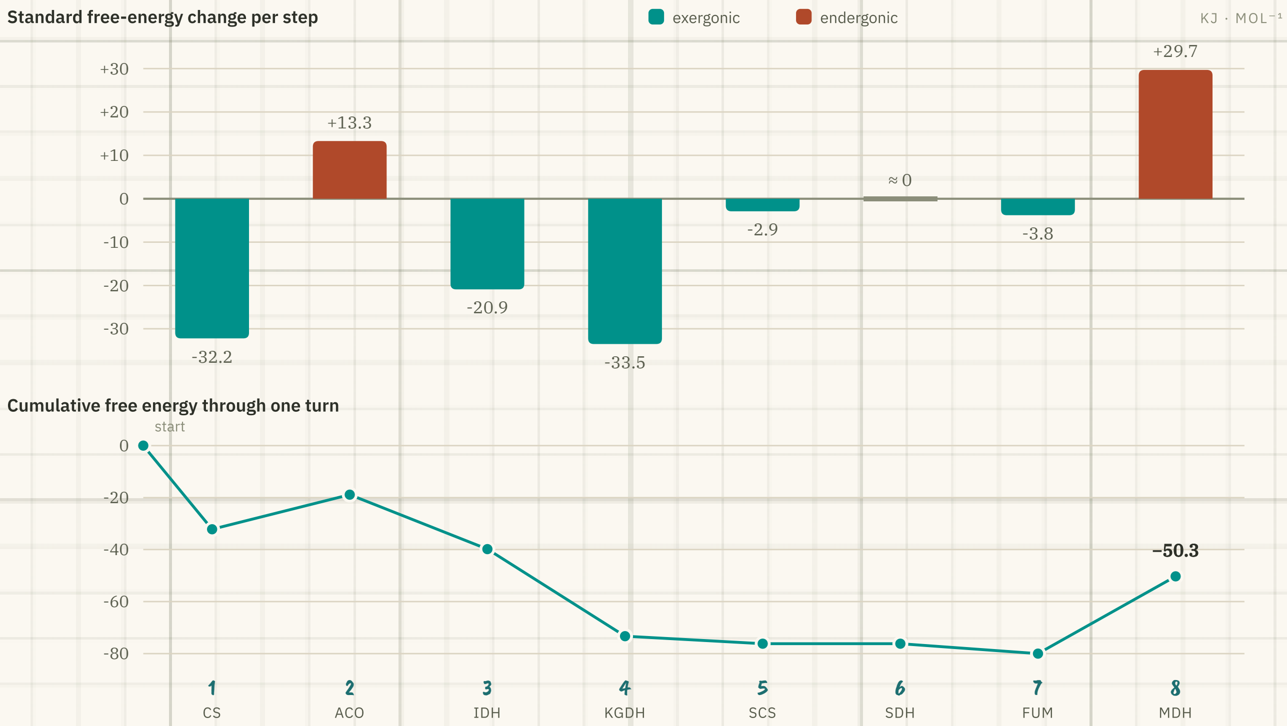
An oxygen ester is stabilised by resonance — the oxygen lone pair delocalises into the carbonyl. Sulfur's 3p orbitals overlap poorly with carbon's 2p, so a **thioester gets almost none of that stabilisation** and hydrolyses with $\Delta G°' \approx$ **−31 kJ·mol⁻¹**, on a par with ATP's −30.5.

The cycle cashes this twice: acetyl-CoA drives the condensation in ***step 1****, and succinyl-CoA is converted straight into GTP in* ***step 5****.*

STEREOSPECIFICITY

Fumarase adds water *anti* across the double bond and accepts only the *trans* isomer. The product is exclusively **L-malate** — maleate, the *cis* isomer, is not a substrate at all.

The energy landscape



Three steps do nearly all the thermodynamic work. Step 8 climbs 29.7 kJ·mol^{-1} back up and still runs forward, because citrate synthase keeps oxaloacetate scarce.

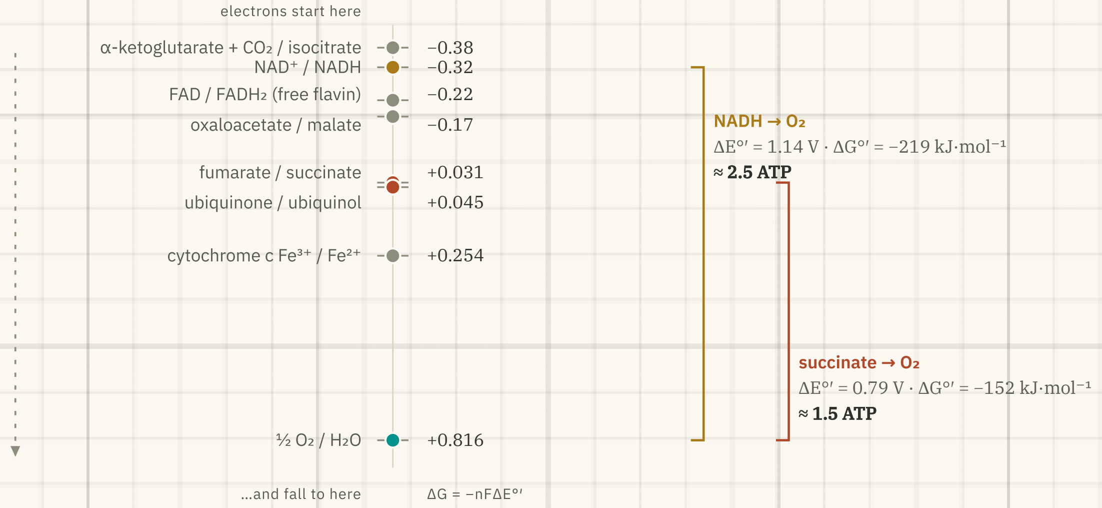
WHAT ΔG°′ DOES AND DOES NOT TELL YOU

Standard values assume 1 M of everything at pH 7 — conditions no cell ever meets. In a working mitochondrion only steps **1, 3 and 4** stay far from equilibrium; the other five sit near ΔG ≈ 0 and run in either direction as concentrations shift.

That is exactly why those three are the regulated steps: a reaction at equilibrium cannot control flux — only a displaced one can.

Why $FADH_2$ is worth less

Standard reduction potential E°' / volts



Electrons only fall from a more negative couple to a more positive one. The size of the fall sets the ATP yield.

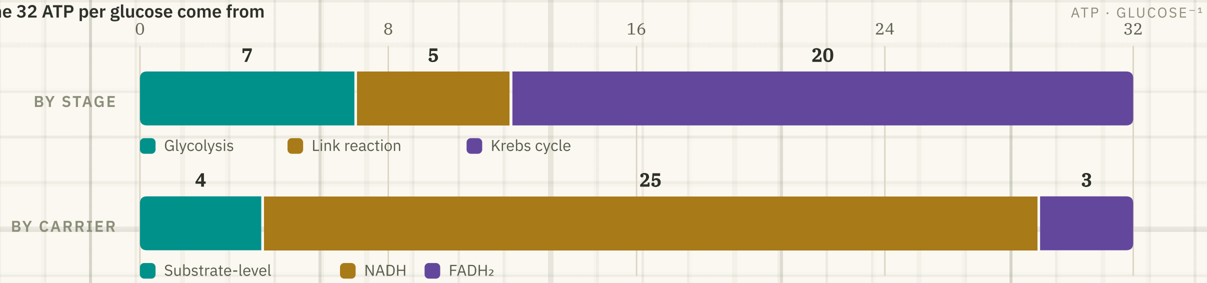
THE ANSWER TO THE QUESTION EVERYONE ASKS

The fumarate/succinate couple sits at **+0.031 V** — about **0.35 V more positive** than NAD^+/NADH. Succinate simply **cannot** reduce NAD^+: the electrons would have to run uphill.

So succinate dehydrogenase uses a tightly bound **FAD** instead and passes its electrons to ubiquinone at +0.045 V — entering the chain *past* Complex I. One proton-pumping site is skipped, and the yield drops from ≈ 2.5 ATP to ≈ 1.5.

Where the ATP comes from

Where the 32 ATP per glucose come from



Liver and heart, using the malate–aspartate shuttle. The same glucose yields 30 ATP in brain and skeletal muscle.

READ BOTH BARS TOGETHER

The cycle contributes **20 of the 32** — but look at the second bar: **25 of the 32** arrive as NADH, and only **4** by direct phosphorylation. Per turn the cycle releases about **50 kJ·mol^{-1}** under standard conditions, barely one ATP's worth of free energy.

Almost everything useful leaves as reduced coenzyme, to be cashed at the respiratory chain. The cycle is the harvest; the chain is the mint.

Three valves, one signal

Unlike glycolysis, the cycle has no single committed step. It is governed by substrate supply and product inhibition at the three irreversible reactions — and every input reduces to one question: *does the cell need energy right now?*

VALVE 1 · STEP 1

Citrate synthase

- ▾ATP, NADH
- ▾succinyl-CoA (product-like)
- ▾citrate (own product)
- ▾long-chain acyl-CoA

Mostly limited by how much acetyl-CoA and oxaloacetate are available.

VALVE 2 · STEP 3 · RATE-LIMITING

Isocitrate dehydrogenase

- ▴ADP, Ca^{2+}
- ▾ATP, NADH

The cycle's true throttle — allosterically sensitive to ADP, so it reads the energy charge directly.

VALVE 3 · STEP 4

α-KG dehydrogenase

- ▴Ca^{2+}
- ▾succinyl-CoA, NADH, ATP

Inhibited by its own product — the same logic as the PDH complex it resembles.

UPSTREAM GATE · COVALENT CONTROL

The PDH complex is switched, not throttled

PDH kinase phosphorylates and **switches it OFF** — activated by ATP, NADH and acetyl-CoA (plenty of fuel already). **PDH phosphatase** dephosphorylates and **switches it ON** — activated by Ca^{2+} and, in adipose tissue, insulin.

THE RULE THAT COVERS ALL OF IT

*High **NADH / NAD^+** and high **ATP / ADP** ⇒ the cycle slows.*

Ca^{2+} — the signal that muscle is contracting — opens three enzymes at once, so ATP supply rises exactly when demand does.

An amphibolic roundabout

Amphibolic = catabolic and anabolic at once. Intermediates are constantly withdrawn for biosynthesis (**cataplerosis**) and must be replaced (**anaplerosis**) or the wheel runs dry.

Citrate
- ↑ exported to cytosol — fatty acid & cholesterol synthesis

α-Ketoglutarate
- ↑ glutamate — amino acids, purines, GABA, glutathione
- ↑ (in) glutamate · glutamine — transamination / glutamate dehydrogenase

Succinyl-CoA
- ↑ δ-aminolaevulinate — porphyrins → haem
- ↑ (in) propionyl-CoA — odd-chain fatty acids, Val · Ile · Met · Thr (B_{12})

Oxaloacetate
- ↑ PEP → gluconeogenesis — aspartate → pyrimidines, urea cycle
- ↑ (in) pyruvate + CO_2 — pyruvate carboxylase, biotin — ★ *the main top-up reaction*

▲ violet — carbon leaves for biosynthesis · ▲ green — carbon is replaced

WHY THIS MATTERS CLINICALLY

Acetyl-CoA cannot replenish oxaloacetate — it only *consumes* it. In starvation or uncontrolled diabetes, oxaloacetate is drained into gluconeogenesis, the cycle cannot keep pace with the acetyl-CoA arriving from fat breakdown, and the surplus is diverted into **ketone bodies**. Hence the old maxim: "fats burn in the flame of carbohydrates."

Counting the ATP

GIVEN	1 glucose fully oxidised in a liver cell (malate–aspartate shuttle). P/O ratios: **2.5 ATP per NADH**, **1.5 ATP per $FADH_2$**.
FORMULA	$ATP_{total} = ATP_{SLP} + 2.5 \cdot n(NADH) + 1.5 \cdot n(FADH_2)$
SUBSTITUTION	Glycolysis — 2 ATP + 2 NADH Link reaction × 2 — 2 NADH Krebs × 2 — 6 NADH + 2 $FADH_2$ + 2 GTP $\mathbf{ATP_{SLP} = 4 \cdot n(NADH) = 10 \cdot n(FADH_2) = 2}$
CALCULATION	$4 + (10 \times 2.5) + (2 \times 1.5) = 4 + 25 + 3$
RESULT	**32 ATP per glucose** — or **30** in brain and skeletal muscle, where the glycerol-3-phosphate shuttle delivers the two cytosolic NADH as $FADH_2$: 4 + 20 + 6.
INTERPRETATION	The cycle itself supplies **20 of the 32 ATP** (≈ 63 %) and **4 of the 6 CO_2** — yet it makes only 2 of them by direct phosphorylation. Its real product is reduced coenzyme; the ATP is minted later, at the electron transport chain.

Where the wheel jams

AGENT	TARGET	CONSEQUENCE
Fluoroacetate (rodenticide 1080)	**aconitase** — step 2	Converted *in the cell* to fluorocitrate, which blocks the enzyme: "lethal synthesis". Citrate accumulates; the cycle halts.
Arsenite & trivalent arsenicals	**lipoamide** — PDH and α-KG dehydrogenase	Binds the two neighbouring thiols of dihydrolipoamide, disabling both complexes at once. Lactate and pyruvate rise.
Malonate	**succinate dehydrogenase** — step 6	Classic competitive inhibitor — the structural analogue of succinate. Krebs used it as the key evidence that the pathway is a cycle.
3-Nitropropionic acid	**succinate dehydrogenase** — irreversible	Selective striatal degeneration; used experimentally to model Huntington disease.
Thiamine (B_1) deficiency	**TPP-dependent enzymes** — PDH, α-KG dehydrogenase, transketolase	Pyruvate cannot enter the cycle → lactic acidosis. Clinically: wet and dry beriberi, Wernicke encephalopathy, Korsakoff syndrome.

Neurons and cardiac myocytes fail first — they have the highest oxidative demand and the least glycolytic reserve.

Intermediates as signals

ONCOMETABOLITES

- **IDH1 / IDH2 mutations** — a neomorphic activity converts α-ketoglutarate into **D-2-hydroxyglutarate**, which blocks α-KG-dependent dioxygenases and drives DNA and histone hypermethylation. Found in gliomas, AML and chondrosarcoma.
- **SDH (A–D) mutations** — succinate accumulates; hereditary paraganglioma, phaeochromocytoma and GIST, with pseudohypoxia through HIF-1α stabilisation.
- **FH mutations** — fumarate accumulates and covalently modifies cysteines ("succination"), including those of aconitase; causes hereditary leiomyomatosis and renal cell carcinoma.

↳ *Three cycle enzymes are bona fide* ***tumour suppressors*** *— metabolism as genetics.*

INHERITED ENZYME DEFECTS

- **Pyruvate dehydrogenase deficiency** — congenital lactic acidosis, hypotonia and neurological impairment; a ketogenic diet helps because ketones enter the cycle downstream of the block.
- **MDH2 mutations** — early-onset encephalopathy, refractory epilepsy and raised blood and CSF lactate.

IMMUNOMETABOLISM

In activated macrophages the cycle is deliberately **broken** at two points so that intermediates accumulate as messengers: succinate stabilises HIF-1α and drives IL-1β, while citrate is diverted to make **itaconate**, which is anti-inflammatory. The cycle is a signalling hub, not just a furnace.

Six things people get wrong

- **"The cycle uses oxygen."** It does not. O_2 is the final electron acceptor at Complex IV; the cycle only depends on that to get NAD^+ and FAD back.
- **"NADH gives 3 ATP, $FADH_2$ gives 2."** Those are the old integer P/O ratios. Current values are **2.5** and **1.5**.
- **"GTP counts as two ATP."** One GTP ≡ one ATP — nucleoside diphosphate kinase interconverts them freely.
- **"The acetyl carbons leave as that turn's CO_2."** They do not — the released carbons come from oxaloacetate.
- **"Acetyl-CoA can be turned into glucose."** Not in humans. Both carbons are lost as CO_2 before oxaloacetate is regenerated, so there is no net gain.
- **"Succinate dehydrogenase sits in the matrix."** It is the one cycle enzyme bound to the inner membrane — it *is* Complex II.

One-glance recap

Everything worth carrying into the exam hall.

KEY NUMBERS

Steps · oxidations · decarboxylations	8 · 4 · 2
Per turn	3 NADH · 1 $FADH_2$ · 1 GTP · 2 CO_2
ATP per turn	10
Turns per glucose	2
ATP per glucose from the cycle	20
ATP per glucose, total	30–32
Rate-limiting enzyme	isocitrate dehydrogenase

FORMULAS WORTH MEMORISING

$$\text{Acetyl-CoA} + 3\,NAD^+ + FAD + GDP + P_i + 2\,H_2O \rightarrow 2\,CO_2 + 3\,NADH + 3\,H^+ + FADH_2 + GTP + \text{CoA-SH}$$

$$ATP = ATP_{SLP} + 2.5\,n(NADH) + 1.5\,n(FADH_2)$$

$$\text{Pyruvate} + \text{CoA-SH} + NAD^+ \rightarrow \text{Acetyl-CoA} + CO_2 + NADH$$

THE THREE VALVES

- **Citrate synthase** — off with ATP, NADH, succinyl-CoA, citrate
- **Isocitrate dehydrogenase** — on with ADP and Ca^{2+}, off with ATP and NADH
- **α-KG dehydrogenase** — on with Ca^{2+}, off with succinyl-CoA, NADH, ATP

DEFINITIONS TO BE ABLE TO WRITE

- **Amphibolic** — serving catabolism and anabolism simultaneously
- **Anaplerotic** — a reaction that replenishes a cycle intermediate; chiefly pyruvate carboxylase
- **Cataplerotic** — withdrawal of an intermediate for biosynthesis
- **Substrate-level phosphorylation** — ATP or GTP made directly from a high-energy substrate, no proton gradient involved
- **Oxidative decarboxylation** — loss of CO_2 coupled to a dehydrogenation (steps 3 and 4)

FINAL TAKEAWAY

The Krebs cycle is not really an ATP factory — it makes just one nucleotide triphosphate per turn. It is an electron harvester and a biosynthetic roundabout: its true output is reduced coenzyme for the respiratory chain, and its intermediates are the raw material for fats, haem, amino acids and glucose. Understand what enters and what leaves, and the whole of metabolism organises itself around this one wheel.

SOURCES CONSULTED

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